

Layering and shear properties of an ionic liquid, 1-ethyl-3-methylimidazolium ethylsulfate, confined to nano-films between mica surfaces

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We report high-resolution measurements of the forces between two atomically smooth solid surfaces across a film of 1-ethyl-3-methylimidazolium ethylsulfate ionic liquid, for film thickness down to a single ion diameter. For films thinner than ~ 2 nm oscillatory structural forces are observed as the surface separation decreases and pairs of ion layers are squeezed out of the film. Strikingly, measurements of the shear stress of the ionic liquid film reveal low friction coefficients which are 1–2 orders of magnitude smaller than for analogous films of non-polar molecular liquids, including standard hydrocarbon lubricants, up to *ca.* 1 MPa pressure. We attribute this to the geometric and charge characteristics of the ionic liquid: the irregular shapes of the ions lead to a low shear stress, while the strong coulombic interactions between the ions and the charged confining surfaces lead to a robust film which is maintained between the shearing surfaces when pressure is applied across the film.

Room temperature ionic liquids (RTILs) are pushing forward the boundaries of possibility in catalysis and synthetic methods,^{1–3} electrochemical applications such as solar cells,^{4–7} lubrication,^{8,9} and nanotechnology.^{10,11} Many of these applications depend pivotally on the interfacial structure and dynamic properties of RTILs, however only in the past few years have these begun to be investigated.^{12–22} We have measured the normal interaction force between atomically smooth solid (mica) surfaces across a confined film of 1-ethyl-3-methylimidazolium ethylsulfate ([EMIM][EtSO₄]), and the shear characteristics of these films as a function of their thickness, down to a single ion layer. Our results reveal strikingly low friction coefficients between the sliding surfaces up to moderate pressures, which we are able to correlate directly with the structure of the film.

Normal and shear forces— F_N and F_S , respectively—were measured as a function of surface separation, D , between two atomically smooth curved mica sheets in crossed-cylinder configuration (radius of curvature $R \approx 1$ cm) across films of [EMIM][EtSO₄] using a surface force balance (SFB). The SFB, custom built for the detection of minute shear stresses, is shown schematically in Fig. 1A and has been described

previously.²³ [EMIM][EtSO₄] was chosen as a model ionic liquid for these experiments because it has a high dielectric constant (thereby increasing the degree of ionisation of the mica surface as described later), a moderate bulk viscosity, and is a common RTIL which has been studied in the past. Experiments were conducted at 25.5 ± 0.3 °C over a period of *ca.* 10 hours. Freshly-cleaved mica sheets of 1.2–1.6 μm thickness and *ca.* 1×1 cm² area were glued to the hemi-cylindrical lenses using epoxy glue (Shell EPON-1004). After calibration of the mica–mica contact in dry air, the [EMIM][EtSO₄] (Solvent Innovation GmbH/Merck KGaA, $\geq 99\%$ purity, Fig. 1B) was injected between the mica surfaces (droplet volume ≈ 100 μL). The [EMIM][EtSO₄] was dried inside the sealed SFB chamber in the presence of P₂O₅ drying agent. Force measurements were reproducible at different contact positions on the mica sheets over the whole period of the experiments (*ca.* 10 hours for each experiment), and in independent experiments using different pairs of mica sheets. The reproducibility of the results over long periods of time, and the presence of clear structural forces at $D < 2$ nm, suggest that there is no effect of surface contamination.

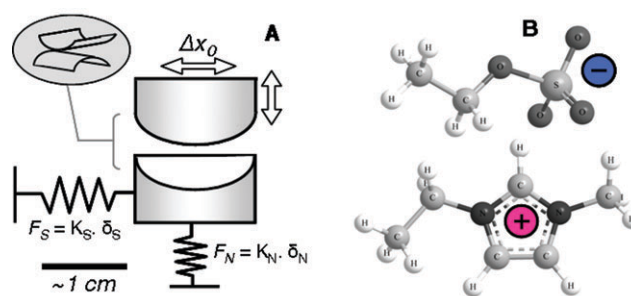


Fig. 1 A: schematic diagram of the SFB setup for normal and shear force measurements. B: ion structures of the [EMIM][EtSO₄].

The normalised equilibrium force law, F_N/R vs. D , across the [EMIM][EtSO₄] film is shown in Fig. 2. F_N is measured from a surface separation of $D \approx 200$ nm (not shown), however no force is detected between the surfaces until they approach closer than 5 nm. When the mica surfaces are brought closer than 5 nm they experience first repulsion then alternate attraction and repulsion, in an oscillatory manner, with increasing magnitude as the separation decreases. This ‘structural force’ originates in the steric interactions between ordered layers of ions at each surface and is in agreement with earlier studies by Horn *et al.*²¹ and by Atkin and Warr.²² Energy minima arise at separations corresponding to favourable ion packing between the surfaces,

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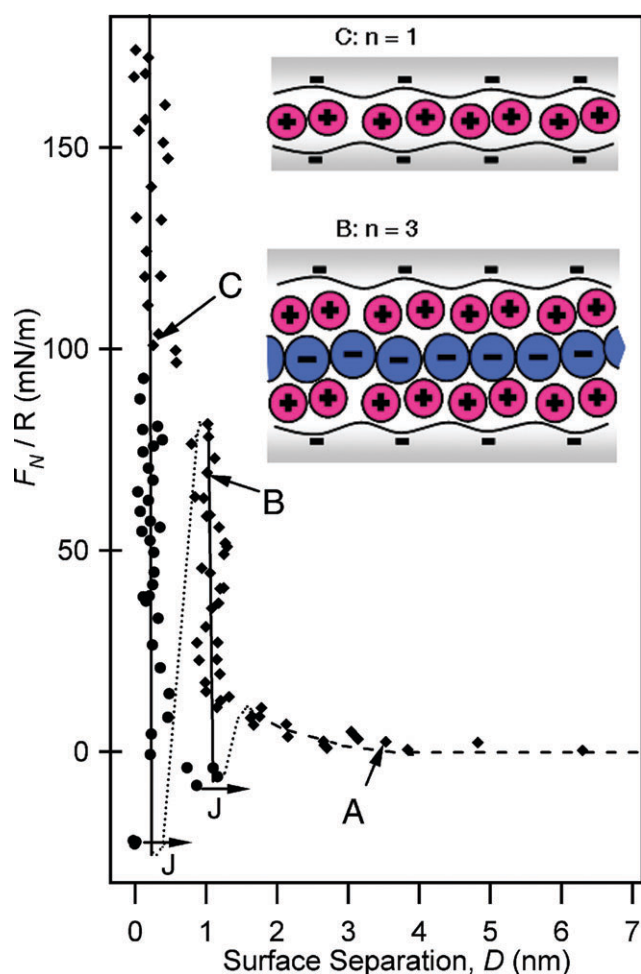


Fig. 2 Interaction force, normalised by radius of curvature, between mica surfaces across [EMIM][EtSO₄] as a function of the surface separation. Diamond data points (◆) were recorded on approach and circles (●) were measured on retraction of the surfaces. The line is a guide to the eye showing the oscillatory nature of the force law. In the region between 1.5 nm and 4.0 nm (dashed line) there may be further oscillations in the force law: this was not investigated in the present study (we focus on the higher pressure behavior), where data points were only collected on approach of the surfaces in this region. Attractive regions are manifested as ‘jumps-in’ due to a spring instability when $dF_N/dD > K_N$ (dotted line regions), at which point RTIL is squeezed out of the film and the surfaces approach to the next closest stable region. Energy minima are similarly detected as jumps-out as shown by arrows. The repulsive regions at 1.08 ± 0.15 nm and 0.23 ± 0.15 nm correspond to three layers (“ $n = 3$ ”) and one layer (“ $n = 1$ ”) of ions between the mica sheets, respectively, as shown in the inset and described in the text.

and energy maxima between are due to disruption of the ordered layering. The closest separation reached is 0.23 ± 0.15 nm (relative to air-contact calibration), which is attributed to a single layer composed predominantly of imidazolium cations (“ $n = 1$ ”) between the negatively charged mica sheets. The adjacent energy minimum at 1.08 ± 0.15 nm is attributed to three layers of ions between the surfaces (“ $n = 3$ ”): a layer of cations at each surface and a central layer containing excess anions. These arrangements are shown schematically in the inset to Fig. 2, and we justify this interpretation as follows.

The mica surfaces are negatively charged due to dissociation of surface K^+ ions into the surrounding RTIL where they are solvated in the bulk liquid (the bulk [EMIM][EtSO₄] has a dielectric constant of $\epsilon = 35.2^{24}$). The negative mica charge is compensated by an excess of [EMIM] cations in the region near each surface:^{14–17} an ‘electrical double layer’. However the double layer in RTILs is likely to be considerably different to the well understood situation in aqueous electrolytes: the ion-correlation effects in these dense coulombic systems are expected to lead to over-compensation of surface charges and oscillations in cation and anion charge density near the charged surface.¹² This has indeed been observed in recent simulations^{14–16} and XRR experiments.¹⁷ In the particular case of mica and imidazolium cations, previous experiments have demonstrated the ion-exchange of K^+ with imidazolium on mica surfaces.²⁵ We have calculated the [EMIM] cation size using both molecular mechanics (CHARMM code) and a semi-empirical package (MOPAC, PM3 method, restricted), each confirming the following dimensions: $(0.76 \pm 0.03 \text{ nm}) \times (0.36 \pm 0.08 \text{ nm}) \times (0.22 \pm 0.04 \text{ nm})$, with the uncertainty arising from the available conformations of the ethyl group. Therefore we propose that between the two mica surfaces at the measured closest energy minimum of 0.23 ± 0.15 nm there exists a layer predominantly composed of [EMIM] cations (an [EMIM] ‘monolayer’ so-called ‘ $n = 1$ ’) lying with aromatic rings parallel to the mica surface. This monolayer has a positive surface charge density which closely compensates the negative surface charge density of the two confining mica sheets, although precise calculations are not meaningful since we do not know the packing orientation of the ions. The next energy minimum (at 1.08 ± 0.15 nm) must include equal numbers of additional cations and anions in order to maintain charge neutrality in the film, and therefore we propose this consists of a monolayer of [EMIM] cations at each negative mica surface and a mid-layer containing an excess of [EtSO₄] anions between (so-called ‘ $n = 3$ ’). This alternate anion-cation interfacial layering picture maintains electro-neutrality in the films, and is in line with recent simulations and experiments.^{14–17} We note that the orientations of the ions and the degrees of exchange or excess between the layers in these films cannot be determined from the present experiments.

The shear properties of the film were characterized by applying a lateral motion to one surface and detecting the shear force, F_S , transmitted across the film to the other surface at different values of surface separation (*i.e.* film thickness). Three examples of this are shown in Fig. 3. F_S remains at or below the resolution of the measurement (*e.g.* Fig. 3A) for $D > 1.5$ nm. When $D = 1.08 \pm 0.15$ nm, corresponding to the $n = 3$ film, a small yield force of $F_{S,y} \approx 5 \mu\text{N}$ is measured (*e.g.* Fig. 3B, and weakly dependent on F_N as described later) above which value the surfaces slide smoothly past one another until the shear direction is reversed. At a separation of $D = 0.23 \pm 0.15$ nm ($n = 1$) the surfaces remain coupled up to higher values of F_S before sliding occurs (Fig. 3C). There is no observable difference between the static ($F_{S,y}$) and the kinetic ($F_{S,k}$) shear force, *i.e.* there is no stiction before sliding or stick-slip during sliding.

In order to study the effect of applied normal load on the friction between the surfaces, we measured the shear forces at

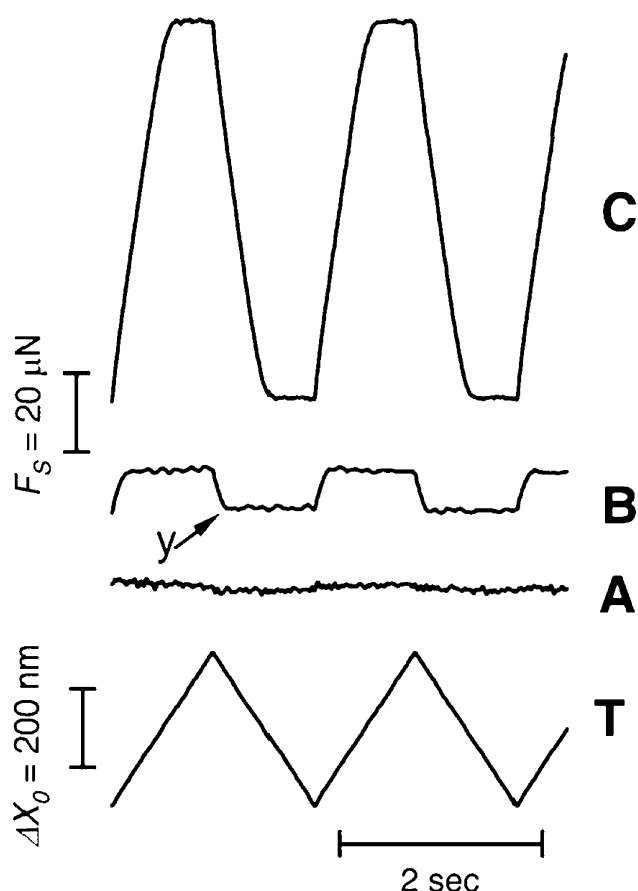


Fig. 3 Examples of the shear force (F_S) response of the confined RTIL films at different values of F_N and D . The applied back-and-forth lateral motion of one mica surface is shown in trace T. Traces A–C, are the response to this applied lateral motion measured at the three points marked A–C, respectively, in Fig. 2. At A ($D \approx 3.5$ nm) F_S stays at or below the noise level. At B ($D \approx 1.08$ nm) a small but measurable shear force is detected: the surfaces are coupled until a yielding value, $F_{S,y}$, is reached (e.g. at the point marked y), then they slide past each other until the shear direction is reversed and the cycle repeats. At C ($D \approx 0.23$ nm) the behavior is similar but with higher yielding shear force. $F_{S,y}$ is half the height between the sliding regimes. The horizontal nature of the sliding regions shows that ‘steady state’ sliding is reached in each cycle before the shear direction is reversed.

different values of F_N and D within the $n = 1$ and $n = 3$ repulsive structural force regions (Fig. 4A). $F_{S,y}$ remains low for the $n = 3$ ($D = 1.08 \pm 0.15$ nm) film, increasing very gently with increasing applied normal force up to ~ 10 μN . From the gradient of this dependence of $F_{S,y}$ on F_N we can determine the coefficient of friction: $\mu_{(n=3)} = dF_S/dF_{N(n=3)} = 0.009 \pm 0.002$. F_S was found to be essentially independent of the rate of applied shear (v/D where v is the shear velocity) over the range studied, as shown in the inset to Fig. 4A. At the point where F_N is sufficient to squeeze out [EMIM][EtSO₄] from the $n = 3$ film, to leave a single layer of imidazolium ions ($n = 1$ at $D = 0.23 \pm 0.15$ nm), F_S increases discontinuously to a higher value (arrows in Fig. 4A). F_S then continues to increase with F_N at a steeper rate compared to the $n = 3$ film, giving $\mu_{(n=1)} = 0.12 \pm 0.02$. Previous macroscopic and AFM studies of lubrication by RTILs have found coefficients of friction in the range $\mu = 0.03$ – 0.8 ,^{8,9} although in those experiments

the pressure and film thickness over the sliding contact were not measured (and are likely to be non-uniform due to surface roughness and asperity contacts). Therefore it is likely that μ -values in those previous studies are average values arising from regions of direct substrate contacts and regions of differing numbers of ion layers and pressures.

Our results demonstrate significantly different shear properties for the RTIL [EMIM][EtSO₄] compared to simple non-ionic liquids such as OMCTS^{23,26} (which shows oscillatory

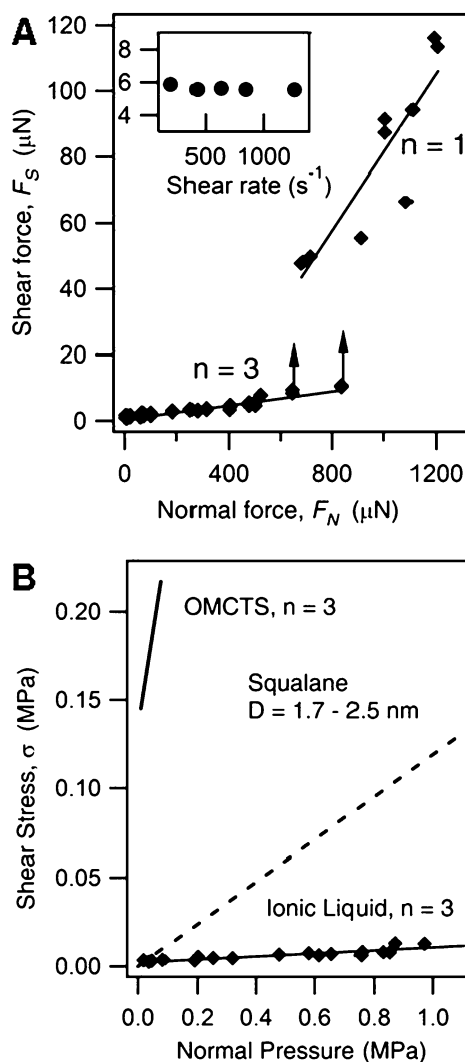


Fig. 4 A: shear force vs. normal force, for the $n = 3$ and $n = 1$ films. Each point is taken from a trace such as in Fig. 3. The inset shows how F_S is essentially independent of shear rate (v/D) over the range investigated (for a fixed value of $F_N \approx 200$ μN). B: comparison of shear stress ($\sigma = F_S/A$; A is the contact area measured directly from the interference pattern for each shear measurement, with typical values of order 10^2 – 10^3 μm^2) vs. normal pressure, P ($P = F_N/A$) for the [EMIM][EtSO₄] and two non-polar molecular liquids octamethylcyclotetrasiloxane (OMCTS; solid line, from ref. 26) and squalane (2,6,10,15,19,23-hexamethyltetracosane, C₃₀H₆₂; dashed line, from ref. 27). All data were measured using a similar SFB method, and are directly comparable to the [EMIM][EtSO₄] $n = 3$ film: the OMCTS also with 3 molecular layers, and the squalane with a similar film thickness of 1.7–2.5 nm. The coefficients of friction, μ , are: $\mu_{(\text{RTIL}, n=3)} = 0.009 \pm 0.002$; $\mu_{(\text{Squalane})} = 0.12$; $\mu_{(\text{OMCTS}, n=3)} = 1.1$.

structural forces qualitatively similar to [EMIM][EtSO₄]), or model lubricant oils such as squalane,²⁷ as shown in Fig. 4B. Firstly, the friction coefficient for [EMIM][EtSO₄] is two orders of magnitude lower than for OMCTS, and one order lower than for squalane, up to a normal pressure of *ca.* 1×10^6 Pa. (At higher pressures the friction coefficient is similar to, slightly less than, that for squalane.) Secondly, the [EMIM][EtSO₄] shows no measurable stick-slip behavior at any shear rate investigated ($195\text{--}1300\text{ s}^{-1}$), in contrast to these and many other non-ionic liquids.

What is the origin of this extreme low shear stress of confined layers of [EMIM][EtSO₄], in particular for film thicknesses down to three ionic layers (*i.e.* up to $\sim 1 \times 10^6$ Pa pressures)? The finite and velocity-independent yield force is characteristic of a solid-like film between the mica surfaces. Nonetheless the magnitude of this yield force and the resulting friction coefficient are extremely low (compared to non-polar liquids or model lubricants), and once this yield point is reached the sliding is smooth. In considering the origin of this lubrication we note that RTILs are ionic materials ‘designed not to solidify’ through their large ion size(s), size mismatch of cation and anion, and non-polar regions on one or both ions.¹ These molecular characteristics lead to bulk liquid properties at room temperature, and often glassy or amorphous (rather than crystalline) solid properties at low temperatures. We propose that these same molecular structural properties are responsible for the low yield force, the smooth (rather than stick-slip) sliding when sheared, and the robust nature of the film under applied load, as follows.

We consider two possible mechanisms which may underlie the low yield stress of an [EMIM][EtSO₄] $n = 3$ film. Firstly, it is possible that at the yield point shear melting occurs, resulting in a liquid-like film during shear. (Albeit a liquid with significantly higher viscosity than the bulk RTIL: the effective shear viscosity, η_{eff} , for the [EMIM][EtSO₄] $n = 3$ film is, $\eta_{\text{eff}} = F_{\text{S,k}}D/Av \approx 10^2\eta_{\text{bulk}}$ where $\eta_{\text{bulk}} = 0.12\text{ Pa s}$; whereas for hydrocarbon liquids confined to films of a few molecular layers the corresponding ratios are typically²³ $\eta_{\text{thin film}}/\eta_{\text{bulk}} \approx 10^4\text{--}10^6$). Alternatively, the film may relieve the stress through slip of one layer across another at the plane of least resistance. Since the shear stress is substantially lower across the $n = 3$ film than for the $n = 1$ film, we can say that this slip is likely to occur predominantly at the [EMIM][EtSO₄] cation layer–anion layer interfaces, rather than at the mica–cation layer interfaces. Once the yield force is reached, we observe smooth sliding even at the lowest shear velocities investigated (180 nm s^{-1}). We suggest that the irregular ion structures—which suppress freezing in the bulk material—lead to disorder of the ions within each layer and therefore an absence of any particularly favourable alignments of adjacent layers, resulting in the low yield force for either melting or slip to occur as well as the absence of stick-slip during sliding.

The charges on the ions of the [EMIM][EtSO₄] appear to play an important role in maintaining the lubricating film between the surfaces as the pressure is increased. The negatively charged surfaces each have a tightly (coulombically) bound layer of cations, which resists ‘squeeze out’ of the film from between the surfaces. Thus a film of [EMIM][EtSO₄] is held between the surfaces up to higher pressures than for

similar films of uncharged molecules (such as OMCTS), resulting in a robust film mediating low friction sliding up to $\sim\text{MPa}$ pressures.

Since the above rationalisation of the data draws upon molecular characteristics which are general to ionic liquids, one is tempted to ask whether or not the results presented here represent generic properties of RTILs. To answer this, however, it will be necessary to investigate a range of RTILs with differing structural and physical characteristics to determine whether this is indeed the case. Such experiments are currently underway.

In summary we have used an SFB with sub-molecular resolution in absolute film thickness and high sensitivity in shear force to measure the shear properties of [EMIM][EtSO₄] films, and to correlate these with the layering properties of the film revealed in the oscillatory normal force profile. We find that molecularly thin films of this RTIL mediate efficient lubrication relative to classic molecular lubricants up to moderate pressures, and that they do not exhibit stiction or stick-slip during shear thereby suppressing potential wear of the sliding surfaces. We attribute these remarkable findings to the combination of charges and irregular ion shapes of the ionic liquid.

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